

Yeshwantrao Chavan College of Engineering, Nagpur

Department of Electronics Engineering

Session 2025-26

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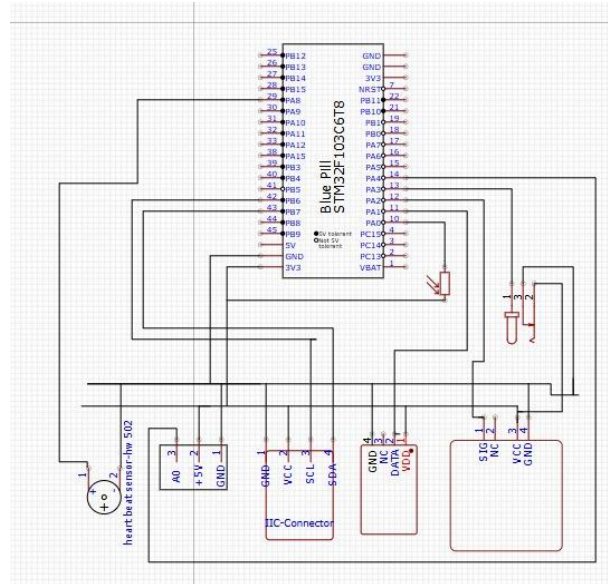
EDA Laboratory Report

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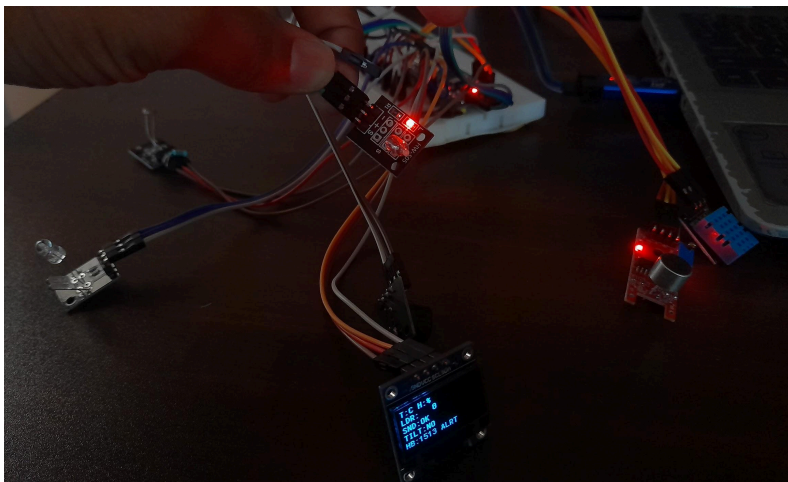
MSPA-04

Title: Smart Health & Environment Monitoring System using STM32 Blue Pill

Circuit Diagram:



Photographs of projects



Ports used:

Port Name	Description of component attached
PA0	LDR Sensor (Analog Input)
PA1	DHT11 Data Pin
PA2	Sound Sensor (Digital Input)
PA3	Tilt Sensor (Digital Input)

PA4	Heartbeat Sensor (Analog Input)
PA8	Buzzer (Output)
PB6	OLED SCL
PB7	OLED SDA

Code with comments:

Code	Comment
HAL_Init();	Initializes the HAL library and resets all peripherals.
SystemClock_Config();	Configures the system clock for STM32 operation.
MX_GPIO_Init();	Initializes GPIO pins for sensors and buzzer.
MX_ADC1_Init();	Initializes ADC1 for analog sensor reading.
MX_I2C1_Init();	Initializes I2C1 for OLED communication.
MX_TIM2_Init();	Initializes TIM2 for microsecond delay used in DHT11.
HAL_TIM_Base_Start(&htim2);	Starts timer 2 for delay generation.
ssd1306_Init();	Initializes the OLED display.
ssd1306_Fill(Black);	Clears the OLED display screen.
ssd1306_SetCursor(10, 10);	Sets cursor position on OLED.
ssd1306_WriteString("MSPA-4 PROJECT", Font_7x10, White);	Displays project title on OLED.
ssd1306_UpdateScreen();	Updates the OLED with new display data.
DHT11_Start();	Sends start signal to DHT11 sensor.
if (DHT11_CheckResponse())	Checks whether DHT11 responds correctly.
Rh_byte1 = DHT11_ReadByte();	Reads humidity integer data from DHT11.
Temp_byte1 = DHT11_ReadByte();	Reads temperature integer data from DHT11.
humidity = (float)Rh_byte1;	Stores humidity value in float format.
temperature = (float)Temp_byte1;	Stores temperature value in float format.
ldrValue = Read_ADC_Channel(ADC_CHANNEL_0);	Reads analog value from LDR sensor on PA0.
heartValue = Read_ADC_Channel(ADC_CHANNEL_4);	Reads analog value from heartbeat sensor on PA4
soundState = HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_2);	Reads digital status of sound sensor.
tiltState = HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_3);	Reads digital status of tilt sensor.
alert = 0;	Resets alert status before checking conditions.

if (temperature > 35.0f) alert = 1;	Sets alert if temperature exceeds threshold.
if (soundState == GPIO_PIN_RESET) alert = 1;	Sets alert if sound is detected.
if (tiltState == GPIO_PIN_RESET) alert = 1;	Sets alert if tilt or movement is detected.
if (heartValue > 3000) alert = 1;	Sets alert if heartbeat raw value is above threshold.
if (alert) Buzzer_On();	Turns buzzer ON during abnormal condition.
else Buzzer_Off();	Turns buzzer OFF during normal condition.
sprintf(msg, "T:%.0fC H:%.0f%%", temperature, humidity);	Formats temperature and humidity data for display.
sprintf(msg, "LDR:%4d", ldrValue);	Formats LDR value for OLED display.
sprintf(msg, "SND:%s", (soundState == GPIO_PIN_RESET) ? "DET" : "OK");	Displays sound sensor status on OLED.
sprintf(msg, "TILT:%s", (tiltState == GPIO_PIN_RESET) ? "YES" : "NO");	Displays tilt sensor status on OLED.
sprintf(msg, "HB:%4d", heartValue);	Formats heartbeat sensor value for display.
Buzzer_On();	Function to activate buzzer.
Buzzer_Off();	Function to deactivate buzzer.
HAL_GPIO_WritePin(GPIOA, GPIO_PIN_8, GPIO_PIN_SET);	Sets PA8 HIGH to turn buzzer ON.
HAL_GPIO_WritePin(GPIOA, GPIO_PIN_8, GPIO_PIN_RESET);	Sets PA8 LOW to turn buzzer OFF.
delay_us(30);	Provides microsecond delay for DHT11 timing.
HAL_TIM_SET_COUNTER(&htim2, 0);	Resets timer counter for precise delay.
while (__HAL_TIM_GET_COUNTER(&htim2) < us);	Waits until required microseconds elapse.
Set_DHT11_Pin_Output();	Configures DHT11 pin as output mode.
Set_DHT11_Pin_Input();	Configures DHT11 pin as input mode.
HAL_Delay(100);	Creates small delay for real-time refresh.

Working:

- Sensors collect real-time data
- STM32 processes using ADC & GPIO
- Values compared with threshold
- If abnormal → buzzer ON
- OLED displays live data

Result:

- Real-time monitoring achieved
- Sensors working properly
- OLED displaying values
- Buzzer giving alert

Conclusion:

The project successfully demonstrates a real-time monitoring system using STM32 Blue Pill. It integrates multiple sensors and provides alert mechanisms, making it useful for health and environmental monitoring applications.